GENEVOLVE
Genomics Division

LPL – PRODUCTION TEST COLLECTION CENTRE
SECTOR - 18, BLOCK-E
ROHINI
DELHI 110085

Name : DUMMY	Collected : 01/06/2021 05:06:00
Lab No. : DUMMYN776 Age: 25 Years Gender: Male	Received : 01/06/2021 17:06:03
	Reported : 04/06/2021 15:31:51
A/c Status : P Ref by : DR. DUMMY DUMMY	Report Status : Final

Nx Gen Seq, Corneal Dystrophy

CLINICAL DETAILS

Individual is a XXX years old XXXX. Next Gen. Sequencing was requested to find any genetic association for Corneal Dystrophy.

FAMILY HISTORY AND PEDIGREE ANALYSIS

RESULT SUMMARY

1 No clinically significant variants were found in relation to phenotypic presentation of the patient.

VARIANT TABLE

GENE	Genomic location	VARIANT*	EXON	TYPE	ZYGOSITY	INHERITANCE	CONDITION/ PHENOTYPE GROUP	CLASSIFICATION*
					NONE			

ANALYSIS STATISTICS

% TARGET NUCLEOTIDES COVERED
≥ 20X
Coverage – 93.36%
Quality Threshold # - 97.43%

#- The quality threshold refers to the percentage of the defined target region of the gene relevant to the given phenotype where read depth was atleast 20x coverage to permit high quality exome variant base calling, annotation and evaluation. Average quality threshold may range from >90-95% of the targeted region, indicating a small portion of the target region may not be covered with sufficient depth or quality to call variant positions confidently.

TEST INFORMATION/LIMITATIONS

Exome sequencing is a targeted sequencing approach that is restricted to the protein-coding regions of genomes. The exome is estimated to encompass approximately 1% of the genome, yet contains approximately 85% of disease-causing mutations [1]. For genetic researchers trying to identify the genes implicated in over 6,800 rare diseases [2], exome sequencing enables

Sample Report

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rapid, cost-effective identification of common single nucleotide variants (SNVs) and small insertions or deletions (indels), as well as rare *de novo* mutations that may explain the heritability of Mendelian and complex disorders [3]. The results are interpreted in the context of the provided clinical findings, family history, and other laboratory data. Only variants in genes potentially related to the proband's medical condition are reported.

The test should not be used for detection of complex genetic events such as copy number variations (CNVs), inversions, translocations and for analysis of sequence repeats or for diagnosis of disorders caused by mutations in the mitochondrial DNA. In addition, due to technology limitations, certain regions may be either not or poorly covered. In these regions variants cannot be confidently detected. Extremely low coverage calls (homo/hemizygous or heterozygous calls with less than three or four reads, respectively) are expected to be artifacts based on our extensive validations and consequently are not considered during the analysis.
Misinterpretation of results may occur if the provided information is inaccurate and/or incomplete. If the obtained genetic results do not concur with the clinical findings, additional testing should be considered.

*VARIANT CLASSIFICATION (BASED ON ACMG RECOMMENDATIONS)

Class 1 – Pathogenic
Class 2 – Likely pathogenic
Class 3 – Variant of uncertain significance (VUS)
Class 4 – Likely benign
Class 5 – Benign
Additionally, other types of clinical relevant variants can be identified (e.g. risk factors, modifiers).

SECONDARY FINDINGS#

Gene	Variant*	Zygosity	Condition group	Significance	Inheritance
			NONE		

#Secondary findings are genetic test results that provide information about changes (variants) in a gene unrelated to the primary purpose for the testing.
In 2013 (and again in 2017), the American College of Medical Genetics and Genomics (ACMG) recommended that all labs performing whole exome and whole genome sequencing tests include the reporting of secondary findings, in addition to any variants that are found related to the primary purpose of the testing. The ACMG proposed a list of 59 genes that are associated with a variety of conditions, from cancer to heart disease. The 59 genes for which secondary findings are reported were chosen because they are associated with conditions that have a definable set of clinical features, the possibility of early diagnosis, a reliable clinical genetic test, and effective intervention or treatment. The goal of reporting these secondary findings to an individual is to provide medical benefit by preventing or better managing health conditions. The variants that are reported are known to cause disease. **Variants of unknown significance**, whose involvement in disease at the current time is unclear, are not reported.

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INCIDENTAL FINDINGS (alterations of unlikely clinical relevance)

None identified.

Incidental Findings are alterations of unlikely clinical relevance, but which were unable to be ruled out. These may include (1) alterations in characterized genes with minimal or inconsistent clinical overlap, (2) a single heterozygous alteration in a recessive gene with limited or inconsistent clinical overlap, or (3) alterations in characterized or uncharacterized genes which failed to co-segregate with disease in the current family. These alterations are not systematically confirmed via Sanger sequencing. These findings are available on request.

TEST METHODOLOGY

DNA was isolated and exome library preparation was performed using the Ion AmpliSeq™ Exome RDY Kit (Thermo Fisher Scientific, Inc.). This kit enables high-efficiency enrichment with **more than 90% target bases covered at 20x and more than 90% uniformity**. The target regions were amplified using the Ion Ampliseq™ Exome RDY Library Preparation.

Template-positive ISPs were enriched, and sequenced on the Ion Proton™ as per the manufacturer's instructions.

VARIANT CALLING AND PRIORITIZATION

Ampliseq exome libraries were sequenced using Ion torrent platform, Proton sequencer. Primary analysis and secondary analysis of the data is performed by Torrent Suite V 5.0.5. Generated data is mapped to the GRCh37/hg19 genome sequence. Several final refinements such as soft-clipping and adapter trimming are routinely performed by torrent suite in the server. Variant calling is performed by an inbuild plugin Variant caller tvc 5.0-13. Annotation of this variant information in done by the Ion Reporter software.

All disease-causing variants reported in HGMD and ClinVar, as well as all variants with minor allele frequency (MAF) below 0.05 in gnomAD database are considered. The investigation for relevant variants is focused on coding exons and UTR regions. All potential modes of inheritance patterns are considered. *In silico* prediction of non-synonymous variants effect is calculated using multiple algorithms such as PolyPhen-2, SIFT, MutationTaster, Mutation Assessor, etc. In addition, provided family history and clinical information are used to evaluate identified variants with respect to their pathogenicity and causality, and are categorized into classes 1 – 5 according to ACMG guidelines. All variants related to the phenotype of the patient, except benign or likely benign variants, are reported.

DISCLAIMER

Any preparation and processing of a sample from patient material provided to GENEVOLVE by a physician, clinical institute or a laboratory (by a "Partner") and the requested genetic and/or biochemical testing itself is based on the highest and most current scientific and analytical standards. However, in very few cases genetic or biochemical tests may not show the correct result, e.g. because of the quality of the material provided by a Partner to GENEVOLVE or in cases where any test provided by GENEVOLVE fails for unforeseeable or unknown reasons that cannot be influenced by GENEVOLVE in advance. In such cases, GENEVOLVE shall not be responsible and/or liable for the incomplete, potentially misleading or even wrong result of any

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testing if such issue could not be recognized by GENEVOLVE in advance.

This report provides information about the patient's mutations that may aid the physician's decision making process, but this test should not be the sole source of information for making decisions on patient care and treatment. These tests should be interpreted in the context of standard clinical, laboratory, and pathological findings. Identification of a mutation in one or more of these genes does not guarantee activity of the drug in a given indication. Insertions and deletions greater than 20bp in size may not be detected by this assay. Mutations in the intronic regions and CNVs have not been included in this report.

The information provided in this report was collected from various sources that we believe to be reliable and quality control procedures have been put in place to ensure the information provided is as accurate, comprehensive, and current as possible. The information provided should only be utilized as a guide or aid and the decision to select any therapy option based on the information reported here resides solely with the discretion of the treating physician. Patient care and treatment decisions should only be made by the physician after taking into account all relevant information available including but not limited to the patient's condition, family history, findings upon examination, results of other diagnostic tests, and the current standards of care. This report should only be used as an aid and the physician should employ clinical judgment in arriving at any decision for patient care or treatment.

RAW DATA

A table with additional variant filtering details can be provided with the raw data (if requested).

REFERENCES

1. Liu Qi, et al. Exome-assistant: a rapid and easy detection of disease-related genes and genetic variations from exome sequencing. BMC Genomics 2012; 13:692.
2. National Institutes of Health, Office of Rare Diseases Research.
3. Marth AR, et al. The functional spectrum of low-frequency coding variation. Genome Biol 2011; 12(9):R84.
4. Hamosh A, et al. Online Mendelian Inheritance in Man (OMIM), a knowledgebase of human genes and genetic disorders. Nucleic Acids Research 2005; 33(DB), D514–D517. http://doi.org/10.1093/nar/gki033, https://www.omim.org/
5. Landrum MJ, et al. ClinVar: public archive of relationships among sequence variation and human phenotype. Nucleic Acids Research 2014; 42 (Database issue), D980–D985. https://www.ncbi.nlm.nih.gov/clinvar/
6. Richards S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015;17(5):405-24.

---- End of report ----

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DMC-28322

Dr Lopamudra Deka
DCP; DNB (Pathology)
Deputy HOD Histopath, Cytopath
Chief of Academic Program – NRL

&

Dr Atul Thatai
PhD, Biotechnology
National Head - Molecular Diagnostics /R & D
NRL - Dr Lal PathLabs Ltd

Sample Report

IMPORTANT INSTRUCTIONS

•Test results released pertain to the specimen submitted. •All test results are dependent on the quality of the sample received by the Laboratory. •Laboratory investigations are only a tool to facilitate in arriving at a diagnosis and should be clinically correlated by the Referring Physician. •Sample repeats are accepted on request of Referring Physician within 7 days post reporting. •Report delivery may be delayed due to unforeseen circumstances. Inconvenience is regretted. •Certain tests may require further testing at additional cost for derivation of exact value. Kindly submit request within 72 hours post reporting. •Test results may show interlaboratory variations. •The Courts/Forum at Delhi shall have exclusive jurisdiction in all disputes/claims concerning the test(s) & or results of test(s). •Test results are not valid for medico legal purposes. *Contact customer care Tel No. +91-11-39885050 for all queries related to test results.
(#) Sample drawn from outside source. (*) Not in NABL scope.